K Non-Riemannian Space

K.1 Metric Connection and Riemann–Cartan Space

In the following subsections, we are interested in four-dimensional space with a metric connection which satisfies metricity condition $\nabla_\rho g_{\mu\nu} = 0$, i.e., a space with $Q_{\mu\nu\rho} = 0$ (or $N^\rho{}_{\nu\mu} = 0$). Hence, the affine connection is expressed by

$$\boxed{\Gamma^\rho{}_{\nu\mu} = \left\{ {\rho \atop \nu\mu} \right\} + K^\rho{}_{\nu\mu}.} \tag{K.1}$$

Linear connection. One can introduce an *anholonomic frame* $\{e_i\}$ which can *not* be written as a form of a coordinate frame $\{\frac{\partial}{\partial x^i}\}$, while the introduced *anholonomic coframe* $\{\vartheta^i\}$ can *not* be done so by an exact 1-form $\{dx^i\}$. A *linear connection 1-form* $\omega^i{}_j = \omega^i{}_{jk}\,\vartheta^k$ can be obtained by the differential of frame fields $De_i = e_j\,\omega^j{}_i$ on a vector bundle E associated with the linear frame bundle $L(\mathcal{M})$ with the *structure group* $\mathrm{GL}(4;\mathbb{R})$. We always use notation $D \equiv D^\omega$ to denote the connection in the *anholonomic frame* in order to distinguish the holonomic one, $\nabla \equiv \nabla^\Gamma$, in the *holonomic frame*.

One can introduce a *nonsingular tetrad* (*vierbein*) $g = (e_i{}^\mu) \in \mathrm{GL}(4;\mathbb{R})$ soldering the anholonomic frame and the holonomic one by $e_i = \partial_\mu \cdot e_i{}^\mu$ (resp. $\vartheta^i = e^i{}_\mu \cdot dx^\mu$). The tetrad field can be regarded as the components (coordinates) of an anholonomic frame written in the holonomic frame, i.e., $\{x^\mu, e_i{}^\mu\}$ is the local coordinate system in the fibre $\pi^{-1}(U) \approx U \times \mathrm{GL}(n;\mathbb{R})$ over the coordinate neighborhood U. The affine connection act on a coframe ϑ^i (a $(0,1)$-type tensor) in local coordinates is

$$\nabla\vartheta^i = \nabla(e^i{}_\mu\,dx^\mu) = de^i{}_\mu \otimes dx^\mu + e^i{}_\mu \nabla dx^\mu = (de^i{}_\mu - e^j{}_\nu\,\Gamma^\nu{}_\mu) \otimes dx^\mu\,. \tag{K.2}$$

On the other hand, the left-hand side of (K.2) can be realized in an anholonomic frame by

$$\nabla\vartheta^i = -\omega^i{}_j \otimes \vartheta^j = (-\omega^i{}_j\,e^j{}_\mu) \otimes dx^\mu\,. \tag{K.3}$$

As a result from the equations given above, we obtain an important equation called the *tetrad-connection equation*

$$\boxed{de^i{}_\mu - e^i{}_\nu\,\Gamma^\nu{}_\mu + \omega^i{}_j\,e^j{}_\mu = 0.}\,^{\dagger 22} \tag{K.4}$$

The equation (K.4) is, in fact, the transformation of a connection induced by the frame transformation. The frame transformation can be given by the form:

$$e_i \longmapsto \partial_\mu = e_i \cdot e^i{}_\mu \quad (\text{resp.} \quad \vartheta^i \longmapsto dx^\mu = e_i{}^\mu \cdot \vartheta^i). \tag{K.5}$$

This is exactly a $\mathrm{GL}(4;\mathbb{R})$ *gauge transformation* and it induces the transformation of a connection form (*gauge potential*):

$$\omega^i{}_j \longmapsto \Gamma^\mu{}_\nu = e_j{}^\mu\,\omega^i{}_j\,e^i{}_\nu + e_i{}^\mu\,de^j{}_\mu. \tag{K.6}$$

[†22] Some people always call this relation the *tetrad postulate*. However, this equation is always valid for all situations and there is no constraint on connection 1-form ω_{ij}, even if the metric is *not compatible*. Therefore, in my point of view, it is improper to be named "postulate." If we restrict that ω_{ij} should be antisymmetric on the LHS of (K.4), it will sometimes *violate* the tetrad postulate, which will be related to the metric compatible condition and explained in Sec. L.

In the matrix forms of $\boldsymbol{g} \equiv (e_i{}^\mu)$, $\mathsf{e} \equiv (e_i)$, $\boldsymbol{\partial} \equiv (\partial_\mu)$ and $\boldsymbol{\Gamma} \equiv (\Gamma^\mu{}_\nu)$ as well as $\boldsymbol{\omega} \equiv (\omega^i{}_j)$, the transformation equations above are read as

$$\begin{cases} \boldsymbol{g} : \mathsf{e} \longmapsto \boldsymbol{\partial} = \mathsf{e}\boldsymbol{g}^{-1}, & \text{(K.7a)} \\ \boldsymbol{g} : \boldsymbol{\omega} \longmapsto \boldsymbol{\Gamma} = \boldsymbol{g}\,\boldsymbol{\omega}\,\boldsymbol{g}^{-1} + \boldsymbol{g}\,d\boldsymbol{g}^{-1} = \boldsymbol{g}(\boldsymbol{\omega} + d)\boldsymbol{g}^{-1}, & \text{(K.7b)} \end{cases}$$

where $\{e_i\}$ stands for the canonical section of $L(\mathcal{M})$. It can be shown that the tetrad-connection equation (K.4) can also be obtained via a projection relation between two covariant derivatives $\nabla_\nu V^\mu := e_i{}^\mu\, D_\nu\, V^i$ where $D_\nu := e^i{}_\nu\, D_i$. From the tetrad-connection equation, we have two formulas for the affine and the linear connection, which are respectively

$$\begin{cases} D_\rho\, e^i{}_{(\mu)} = \partial_\rho e^i{}_{(\mu)} + \omega^i{}_{j\rho}\, e^j{}_{(\nu)} = e^i{}_\nu\, \Gamma^\nu{}_{\mu\rho} & \text{(K.8a)} \\ \nabla_\rho\, e^{(i)}{}_\mu = \partial_\rho e^{(i)}{}_\mu - \Gamma^\nu{}_{\mu\rho}\, e^{(i)}{}_\nu = -\, e^j{}_\mu\, \omega^i{}_{j\rho} & \text{(K.8b)} \end{cases}$$

with $\omega^i{}_{j\rho} := \omega^i{}_{jk}\, e^k{}_\rho$, respectively. Here the indices in the parenthesis indicate that the indices are not recognized by the connection.

We can also define a *total* connection $\boldsymbol{\nabla}$ with respect to both the holonomic and anholonomic bases. Through $T\mathcal{M}$ and E, we can respectively construct two tensor bundles over $T\mathcal{M}$ and E

$$V^r_s = \bigcup_{p\in\mathcal{M}} \overset{r}{\otimes} T_p\mathcal{M} \overset{s}{\otimes} T^*_p\mathcal{M} = \overset{r}{\otimes} T\mathcal{M} \overset{s}{\otimes} T^*\mathcal{M} \tag{K.9a}$$

$$W^k_\ell = \bigcup_{p\in\mathcal{M}} \overset{k}{\otimes} E_p \overset{\ell}{\otimes} E^*_p = \overset{k}{\otimes} E \overset{\ell}{\otimes} E^*, \tag{K.9b}$$

where $T_p\mathcal{M}$ is a tangent space at $p \in \mathcal{M}$ and E_p is a vector space at $p \in \mathcal{M}$ on which is acted by the structure group $\mathrm{GL}(4, \mathbb{R})$ of $L(\mathcal{M})$. By V^r_s and W^k_ℓ, we can obtain a *product bundle*

$$V^r_s \otimes W^k_\ell = \overset{r}{\otimes} T\mathcal{M} \overset{s}{\otimes} T^*\mathcal{M} \overset{k}{\otimes} E \overset{\ell}{\otimes} E^*. \tag{K.10}$$

Therefore the connection defined on $V^r_s \otimes W^k_\ell$ gives

$$\boldsymbol{\nabla} : \Gamma(V^r_s \otimes W^k_\ell) \longrightarrow \Gamma(V^r_{s+1} \otimes W^k_\ell). \tag{K.11}$$

e.g., Let $\{\partial_\mu\}$ and $\{e_i\}$ be the frame of $T\mathcal{M}$ and E respectively. Coframes $\{dx^\mu\}$ and $\{\vartheta^i\}$ are respectively their corresponding dual frames of $\{\partial_\mu\}$ and $\{e_i\}$. We assume that $\Gamma^\mu{}_\nu$ is the connection form on $T\mathcal{M}$, while $\omega^i{}_j$ is the connection form on E. Given $X = X^\mu{}_{\nu\rho}{}^i{}_j\, \partial_\mu \otimes dx^\nu \otimes dx^\rho \otimes e_i \otimes \vartheta^j \in \Gamma(V^1_2 \otimes W^1_1)$, its covariant derivative is

$$\boldsymbol{\nabla} X = \left(\boldsymbol{\nabla}_\sigma X^\mu{}_{\nu\rho}{}^i{}_j\right) \partial_\mu \otimes dx^\sigma \otimes dx^\nu \otimes dx^\rho \otimes e_i \otimes \vartheta^j, \tag{K.12}$$

where

$$\begin{aligned} \boldsymbol{\nabla}_\sigma X^\mu{}_{\nu\rho}{}^i{}_j = {} & \partial_\sigma X^\mu{}_{\nu\rho}{}^i{}_j + \Gamma^\mu{}_{\lambda\sigma}\, X^\lambda{}_{\nu\rho}{}^i{}_j - \Gamma^\lambda{}_{\nu\sigma}\, X^\mu{}_{\lambda\rho}{}^i{}_j \\ & - \Gamma^\lambda{}_{\rho\sigma}\, X^\mu{}_{\nu\lambda}{}^i{}_j + \omega^i{}_{k\sigma}\, X^\mu{}_{\nu\rho}{}^k{}_j + \omega^k{}_{j\sigma}\, X^\mu{}_{\nu\rho}{}^i{}_k \in \Gamma(V^1_3 \otimes W^1_1). \end{aligned} \tag{K.13}$$

Metric. In Riemann–Cartan space U_4, we can introduce a fundamental tensor, i.e., a *metric tensor* $g = g_{\mu\nu}\, dx^\mu \otimes dx^\nu$. In *anholonomic frame*, the metric tensor has form $g = g_{ij}\, \vartheta^i \otimes \vartheta^j$. In particular, if we choose a particular anholonomic (*psueo-*)*orthonormal* frame such that the metric tensor $g(e_i, e_j) = g_{ij} = g_{\mu\nu} e_i{}^\mu e_j{}^\nu$ where g_{ij} is read by the *Euclidean* metric tensor $g_{ij} = \delta_{ij}$ (resp. the physical spacetime described by the *Minkowski* metric tensor $g_{ij} = \eta_{ij}$), and the corresponding connection D is especially called the *Euclidean connection* (resp. the *Lorentzian connection*). Then the structure group should be restricted to SO(4) (resp. SO(1, 3)) $\subset$ GL(4; $\mathbb{R}$), and then a vector bundle E associated with a principal subbundle $Q(\mathcal{M}) \subset L(\mathcal{M})$ over $\mathcal{M}$ can be constructed. On the vector bundle E the transformation of SO(4) (resp. SO(1, 3)) should be

$$\begin{cases} e_i \longmapsto e'_i = e_j (A^{-1})^j{}_i \\ V^i \longmapsto V'^i = A^i{}_j V^j \end{cases} \quad \text{with} \quad g = (A^i{}_j) \in \mathrm{SO}(4) \quad \Big(\text{resp.} \quad g = (\Lambda^i{}_j) \in \mathrm{SO}(1,3)\Big). \tag{K.14}$$

Metric connection. The most important property of the metric tensor is

$$\begin{aligned} \nabla g &= \nabla\big(g_{\mu\nu} dx^\mu \otimes dx^\nu\big) \\ &= \big(dg_{\mu\nu}\big) dx^\mu \otimes dx^\nu + g_{\mu\nu}\big(\nabla dx^\mu\big) \otimes dx^\nu + g_{\mu\nu} dx^\mu \otimes \big(\nabla dx^\nu\big) \\ &= \big(dg_{\mu\nu} - \Gamma_{\nu\mu} - \Gamma_{\mu\nu}\big)\, dx^\mu \otimes dx^\nu = 0\,, \end{aligned} \tag{K.15}$$

which is called *metric compatible condition* or *metricity condition* and the corresponding connection is called the *metric connection*. For an *anholonomic* frame,

$$0 = Dg = \big(\underbrace{dg_{ij}}_{0} - g_{kj}\, \omega^k{}_i - g_{ik}\, \omega^k{}_j\big)\, \vartheta^i \otimes \vartheta^j = \big(-\, \omega_{ji} - \omega_{ij}\big)\, \vartheta^i \otimes \vartheta^j\,. \tag{K.16}$$

The metric compatible condition apparently gives the antisymmetric property of the connection in the *anholonomic* frame, i.e., $\omega_{ij} = \omega_{[ij]}$.

Under the restriction of E associated to the principal bundle $Q(\mathcal{M}) \subset L(\mathcal{M})$ with the structure group SO(4) (resp. SO(1, 3)) $\subset$ GL(4; $\mathbb{R}$) over $\mathcal{M}$, the *metric compatible* condition for *total* covariant derivative $\boldsymbol{\nabla} g = 0$ (or, more precisely, $\boldsymbol{\nabla}_\rho\, g_{\mu\nu} = 0$) is equivalent to have two following conditions:

- Tetrad-connection relation $\boldsymbol{\nabla} e = 0$ (or more precisely, $\boldsymbol{\nabla}_\rho\, e^i{}_\mu = \partial_\rho\, e^i{}_\mu - e^i{}_\nu\, \Gamma^\nu{}_{\mu\rho} + \omega^i{}_{j\rho}\, e^j{}_\mu = 0$), where $e \coloneqq e^i{}_\mu\, e_i \otimes dx^\mu \in \Gamma(E \otimes T^*\mathcal{M})$. This is, indeed, the equation (K.4).
- *Anti-symmetry* condition of the Euclidean (resp. Lorentzian) connection $\omega_{ij\rho} = -\omega_{ji\rho}$.

It can be shown by

$$\begin{aligned} 0 = \boldsymbol{\nabla}_\rho\, g_{\mu\nu} &= \boldsymbol{\nabla}_\rho\big(g_{ij}\, e^i{}_\mu\, e^j{}_\nu\big) \\ &= \big(\boldsymbol{\nabla}_\rho\, g_{ij}\big) e^i{}_\mu\, e^j{}_\nu + g_{ij}\big(\underbrace{\boldsymbol{\nabla}_\rho\, e^i{}_\mu}_{0}\big) e^j{}_\nu + g_{ij}\, e^i{}_\mu \big(\underbrace{\boldsymbol{\nabla}_\rho\, e^j{}_\nu}_{0}\big) \\ &= \big(\underbrace{\partial_\rho g_{ij}}_{\partial_\rho \delta_{ij} = 0} - g_{kj}\, \omega^k{}_{i\rho} - g_{ik}\, \omega^k{}_{j\rho}\big) e^i{}_\mu\, e^j{}_\nu \\ &= -\big(\underbrace{\omega_{ji\rho} + \omega_{ij\rho}}_{0}\big) e^i{}_\mu\, e^j{}_\nu\,. \end{aligned} \tag{K.17}$$

We note that the tetrad-connection equation $\partial_\mu e^i{}_\nu - e^i{}_\rho \Gamma^\rho{}_{\nu\mu} + e^j{}_\nu \omega^i{}_{j\mu} = 0$ is independent with the metricity condition unless the total covariant derivatives is defined. It is induced by the frame transformation and shows the relation between the affine and the Lorentz connection

$$\begin{cases} \Gamma^\rho{}_{\nu\mu} = e_i{}^\rho \, \partial_\mu e^i{}_\nu + e_i{}^\rho \, \omega^i{}_{j\mu} \, e^j{}_\nu = e_i{}^\rho \, D_\mu e^i{}_\nu, & \text{(K.18a)} \\ \omega^i{}_{j\mu} = e^i{}_\nu \, \partial_\mu e_j{}^\nu + e^i{}_\rho \, \Gamma^\rho{}_{\nu\mu} \, e_j{}^\nu = e^i{}_\nu \, \nabla_\mu e_j{}^\nu. & \text{(K.18b)} \end{cases}$$

In terms of the matrix forms by identifying $\mathsf{e} := (e_i{}^\rho)$ and $\mathsf{e}^{-1} := (e^j{}_\nu)$, equations above becomes

$$\begin{cases} \boldsymbol{\Gamma} = \mathsf{e} D \mathsf{e}^{-1}, & \text{(K.19a)} \\ \boldsymbol{\omega} = \mathsf{e}^{-1} \nabla \mathsf{e}. & \text{(K.19b)} \end{cases}$$

In V_4 space, we also have the same relation in the tetrad formulation

$$\left\{ {\rho \atop \nu\mu} \right\} = e_i{}^\rho \, \partial_\mu e^i{}_\nu + e_i{}^\rho \, \Delta^i{}_{j\mu} \, e^j{}_\nu = e_i{}^\rho \, \widetilde{D}_\mu e^i{}_\nu \,, \tag{K.20}$$

where

$$\left\{ {\rho \atop \nu\mu} \right\} = \frac{1}{2} g^{\rho\sigma} \Big(\partial_\nu g_{\sigma\mu} + \partial_\mu g_{\nu\sigma} - \partial_\sigma g_{\nu\mu} \Big) \tag{K.21}$$

is the *Levi–Civita connection* determined by g only. The operation $\widetilde{D}_\mu$ is the *torsion-free* covariant derivative which acts in the anholonomic frame with respect to the Lorentz connection $\Delta^i{}_{j\mu} := \Delta^i{}_{jk} \, e^k{}_\mu$ with the *Ricci rotation coefficients* $\Delta^i{}_{jk} = \Delta^i{}_{jk}(e)$ as a function of tetrad only in GR. The *torsion-free* Lorentz connection is sometimes denoted by $\widetilde{\omega}^i{}_{j\mu}$. We always use the *wide-tilde symbol* to denote the torsion-free quantities.

Now we define the *torsion tensor* $S^\rho{}_{\nu\mu}$ or $T^\rho{}_{\nu\mu}$ which is the *antisymmetric part* of the affine connection

$$\begin{aligned} S^\rho{}_{\nu\mu} &\equiv -\frac{1}{2} T^\rho{}_{\nu\mu} = \frac{1}{2} T^\rho{}_{\mu\nu} \\ &:= \frac{1}{2} \Big(\Gamma^\rho{}_{\nu\mu} - \Gamma^\rho{}_{\mu\nu} \Big) = \Gamma^\rho{}_{[\nu\mu]} = e_i{}^\rho \, \partial_{[\mu} e^i{}_{\nu]} + e_i{}^\rho \omega^i{}_{j[\mu} e^j{}_{\nu]} \,, \end{aligned} \tag{K.22}$$

and $S^\rho{}_{\nu\mu} = -S^\rho{}_{\mu\nu}$. The linear combination of torsion tensors defines a *contortion* tensor in (K.1):

$$\begin{aligned} K^\rho{}_{\nu\mu} &= S^\rho{}_{\nu\mu} - S_\nu{}^\rho{}_\mu - S_\mu{}^\rho{}_\nu \\ &= S^\rho{}_{\nu\mu} + S_{\nu\mu}{}^\rho + S_{\mu\nu}{}^\rho \\ &= S^\rho{}_{\nu\mu} + 2 \, S_{(\nu\mu)}{}^\rho \equiv K^\rho{}_{[\nu\mu]} + K^\rho{}_{(\nu\mu)} \,. \end{aligned} \tag{K.23}$$

where we have $K^\rho{}_{\nu\mu} = -K_\nu{}^\rho{}_\mu$ and $K^\rho{}_{[\nu\mu]} = S^\rho{}_{\nu\mu}$ as well as $K^\rho{}_{(\nu\mu)} = 2 \, S_{(\nu\mu)}{}^\rho$. Alternatively it can be written as

$$\begin{aligned} K^\rho{}_{\nu\mu} &= -\frac{1}{2} \Big(T^\rho{}_{\nu\mu} - T_\nu{}^\rho{}_\mu - T_\mu{}^\rho{}_\nu \Big) \\ &= -\frac{1}{2} \Big(T^\rho{}_{\nu\mu} + T_{\nu\mu}{}^\rho + T_{\mu\nu}{}^\rho \Big) \\ &= -\frac{1}{2} T^\rho{}_{\nu\mu} - T_{(\nu\mu)}{}^\rho \equiv K^\rho{}_{[\nu\mu]} + K^\rho{}_{(\nu\mu)} \,. \end{aligned} \tag{K.24}$$

Particularly we have a *trace relation* between torsion and contortion tensors

$$K^{\mu}{}_{\nu\mu} = -\frac{1}{2}\Bigg(\underbrace{T^{\mu}{}_{\nu\mu}}_{-T^{\mu}{}_{\mu\nu} := -T_{\nu}} - \overbrace{T_{\nu}{}^{\mu}{}_{\mu}}^{0} \underbrace{-T_{\mu}{}^{\mu}{}_{\nu}}_{-T_{\nu}}\Bigg) = T_{\nu} \tag{K.25a}$$

$$\overset{\text{or}}{=} \underbrace{S^{\mu}{}_{\nu\mu}}_{-S^{\mu}{}_{\mu\nu} := -S_{\nu}} - \overbrace{S_{\nu}{}^{\mu}{}_{\mu}}^{0} \underbrace{-S_{\mu}{}^{\mu}{}_{\nu}}_{-S_{\nu}} = -2S_{\nu}\,. \tag{K.25b}$$

In addition, The trace of K can be chosen by the alternative form which should be

$$K^{\nu}{}_{\nu\mu} = -\frac{1}{2}\Bigg(\underbrace{T^{\nu}{}_{\nu\mu}}_{\cancel{T_{\mu}}} \overbrace{-T_{\nu}{}^{\nu}{}_{\mu}}^{\cancel{-T_{\mu}}} - \underbrace{T_{\mu}{}^{\nu}{}_{\nu}}_{0}\Bigg) = 0. \tag{K.26}$$

Consequently, we have the following contractions of an affine connection

$$\begin{cases} \Gamma^{\mu}{}_{\nu\mu} = \left\{{\mu \atop \nu\mu}\right\} + K^{\mu}{}_{\nu\mu} = \left\{{\mu \atop \nu\mu}\right\} + T_{\nu} \overset{\text{or}}{=} \left\{{\mu \atop \nu\mu}\right\} - 2\,S_{\nu}, & \text{(K.27a)} \\ \Gamma^{\nu}{}_{\nu\mu} = \left\{{\nu \atop \nu\mu}\right\} + K^{\nu}{}_{\nu\mu} = \left\{{\nu \atop \nu\mu}\right\}. & \text{(K.27b)} \end{cases}$$

So we obtain an modification equation of (6.128)

$$\partial_{\mu}\sqrt{-\mathbf{g}} = \sqrt{-\mathbf{g}}\,\underbrace{\left\{{\nu \atop \nu\mu}\right\}}_{\left\{{\nu \atop \mu\nu}\right\}} = \sqrt{-\mathbf{g}}\big(\Gamma^{\nu}{}_{\mu\nu} - T_{\mu}\big), \tag{K.28}$$

which results in the *modified covariant divergence*

$$\boxed{\nabla_{\mu}V^{\mu} = \partial_{\mu}V^{\mu} + \frac{\partial_{\mu}\sqrt{-\mathbf{g}}}{\sqrt{-\mathbf{g}}}\,V^{\mu} + T_{\nu}\,V^{\nu} = \frac{1}{\sqrt{-\mathbf{g}}}\,\partial_{\mu}\big(\sqrt{-\mathbf{g}}V^{\mu}\big) + T_{\mu}\,V^{\mu}.} \tag{K.29}$$

The affine connection can always be decomposed as $\Gamma^{\rho}{}_{\nu\mu} = \left\{{\rho \atop \nu\mu}\right\} + K^{\rho}{}_{\nu\mu}$. Under the decomposition of the connection

$$\begin{aligned} \Gamma^{\rho}{}_{\nu\mu} &= \left\{{\rho \atop \nu\mu}\right\} + K^{\rho}{}_{\nu\mu} \\ &= \Big(e_{i}{}^{\rho}\,\partial_{\mu}\,e^{i}{}_{\nu} + e_{i}{}^{\rho}\Delta^{i}{}_{j\mu}e^{j}{}_{\nu}\Big) + e_{i}{}^{\rho}K^{i}{}_{j\mu}e^{j}{}_{\nu} \\ &= e_{i}{}^{\rho}\,\partial_{\mu}\,e^{i}{}_{\nu} + e_{i}{}^{\rho}\Big(\Delta^{i}{}_{j\mu} + K^{i}{}_{j\mu}\Big)e^{j}{}_{\nu} \\ &:= e_{i}{}^{\rho}\,\partial_{\mu}\,e^{i}{}_{\nu} + e_{i}{}^{\rho}\,\omega^{i}{}_{j\mu}\,e^{j}{}_{\nu}\,, \end{aligned} \tag{K.30}$$

we then have a corresponding decomposition of the Euclidean connection (resp. the Lorentz connection) in the anholonomic frame

$$\boxed{\omega^{i}{}_{jk} = \Delta^{i}{}_{jk} + K^{i}{}_{jk}.} \tag{K.31}$$

We note that the contortion in the anholonomic frame is read by

$$\begin{aligned} K_{kji} := g_{k\ell}\,K^{\ell}{}_{ji} &= g_{k\ell}\Big(e^{\ell}{}_{\rho}\,K^{\rho}{}_{\nu\mu}\,e_{j}{}^{\nu}\,e_{i}{}^{\mu}\Big) \\ &= g_{k\ell}\Big(e^{\ell}{}_{\rho}\,(S^{\rho}{}_{\nu\mu} + 2\,S_{(\nu\mu)}{}^{\rho})\,e_{j}{}^{\nu}\,e_{i}{}^{\mu}\Big) \\ &= g_{k\ell}\Big(S^{\ell}{}_{ji} + 2\,S_{(ji)}{}^{\ell}\Big)\,. \end{aligned} \tag{K.32}$$

K.2 Cartan's Structure Equations

Anholonomity and structure constants. The *structure constants* $c^k{}_{ij}$ can be calculated by the *Lie bracket* (commutator) of the anholonomic frame on $T\mathcal{M}$

$$\begin{aligned}
[e_i\,,\,e_j] &= [e_i{}^\mu\,\partial_\mu\,,\,e_j{}^\nu\,\partial_\nu] \\
&= e_i{}^\mu\,\partial_\mu(e_j{}^\nu\,\partial_\nu) - e_j{}^\nu\,\partial_\nu(e_i{}^\mu\,\partial_\mu) \\
&= [e_i{}^\nu(\partial_\nu\,e_j{}^\mu) - e_j{}^\nu(\partial_\nu\,e_i{}^\mu)]\,\underbrace{\partial_\mu}_{e^k{}_\mu\,e_k} \\
&= e_k\left[-e_i{}^\nu e_j{}^\mu(\partial_\nu\,e^k{}_\mu) + e_j{}^\nu e_i{}^\mu(\partial_\nu\,e^k{}_\mu)\right] \\
&= e_k\left[(e^k{}_{\mu,\nu} - e^k{}_{\nu,\mu})\,e_i{}^\mu e_j{}^\nu\right] \\
&\coloneqq e_k\,c^k{}_{ij} = -e_k\,c^k{}_{ji}.
\end{aligned} \tag{K.33}$$

It can be shown that $c^k{}_{ij}$ can also be obtained by the exterior differentiation of a canonical 1-form

$$\begin{aligned}
\mathrm{d}\vartheta^k &= \mathrm{d}(e^k{}_\mu\,dx^\mu) = \partial_\nu e^k{}_\mu\,dx^\nu \wedge dx^\mu \\
&= -e^k{}_{[\mu,\,\nu]}\,dx^\mu \wedge dx^\nu = -(e^k{}_{[\mu,\,\nu]}\,e_i{}^\mu e_j{}^\nu)\,\vartheta^i \wedge \vartheta^j \equiv -\frac{1}{2}\,c^k{}_{ij}\,\vartheta^i \wedge \vartheta^j\,,
\end{aligned} \tag{K.34}$$

which is a *Maurer-Cartan form* of $\mathfrak{gl}(4;\mathbb{R})$. As a result, the structure constants can be written by tetrad field

$$\boxed{c^k{}_{ij} = 2\,e^k{}_{[\mu,\,\nu]}\,e_i{}^\mu e_j{}^\nu.} \tag{K.35}$$

It is apparent that $c^k{}_{ij}$ should be *antisymmetric* in i and j. Due to the Poincaŕe theorem $d\vartheta^k = 0$ if ϑ^k is *locally* exact (i.e., $\vartheta^k = dx^k$ is a *total differential*). This means that all the structure constants $c^k{}_{ij}$ are vanishing and ϑ represents a holonomic frame. Therefore, the structure constant $c^k{}_{ij}$ measures the *anholonomity* of a frame called the *object of anholonomity*.

Exterior covariant differentiation. The *exterior differentiation* of differential forms is a map

$$\mathrm{d} : A^r(\mathcal{M}) \longrightarrow A^{r+1}(\mathcal{M})\,. \tag{K.36}$$

The exterior differentiation of differential forms can be extended to the *exterior covariant differentiation* of *vector-valued r-forms*, which is defined by

$$\mathbf{d} : A^r(E) \longrightarrow A^{r+1}(E)\,,^{\dagger 23} \tag{K.37}$$

where the product bundle $A^r(E) \coloneqq \Gamma(E) \otimes A^r(\mathcal{M}) \overset{\text{or}}{=} \Gamma(E \otimes \Lambda^r T^*\mathcal{M})$ in which $\Gamma(E)$ is a vector space and $A^r(\mathcal{M})$ is a vector space of all the differential form of degree r. Let $e_i \in A^0(E) = \Gamma(E) \otimes A^0(\mathcal{M}) = \Gamma(E)$ be a basis vector and $\vartheta^i \in A^1(\mathcal{M})$ be a basis 1-form. Applying the exterior covariant differentiation, we have

$$\boxed{\begin{cases} \mathbf{d}e_i = \boldsymbol{d}e_i = \nabla e_i = e_j \otimes \omega^j{}_i \in A^1(E), & \text{(K.38a)} \\ \mathbf{d}\vartheta^i = \mathrm{d}\vartheta^i \in A^2(\mathcal{M}). & \text{(K.38b)} \end{cases}}$$

[†23] In stead of the notation $\mathbf{d}$, we can also use d_∇ (or d^∇, or d^ω, or d_ω) to distinguish the *covariant* exterior differentiation from the *ordinary* exterior differentiation. We note that traditionally the symbol is denoted by D, which can be found in books of Kobayashi and Nomizu [10], Spivak [14], etc. Originally Cartan denoted the covariant exterior differentiation by a *prime* and get the notation (r-form)$' \equiv$ d(r-form) and (vector)$' \equiv \nabla$(vector).

So we can identify the covariant exterior derivative by

$$\boxed{\mathbf{d} = (\nabla \otimes \mathrm{id}_{A^r(\mathcal{M})}) \oplus (\mathrm{id}_{\Gamma(E)} \otimes \mathrm{d}).} \tag{K.39}$$

We note that a connection ∇ is called *flat* if the curvature is *free*, i.e., space with $R^\rho{}_{\sigma\mu\nu} = 0$ There are two possibilities:

- for *trivial* case, there exists a global *fixed* frame (e.g., a Cartesian coordinate frame) such that the vectorial variation of a basis vector is zero, i.e., $\mathbf{d}e_i = 0$ with a set of *vanishing* connections $\{\omega^i{}_j\}$;
- for *non-trivial* case, there exists a *non-vanishing* connection $\omega^i{}_j$ such that $R^\rho{}_{\sigma\mu\nu} = 0$.

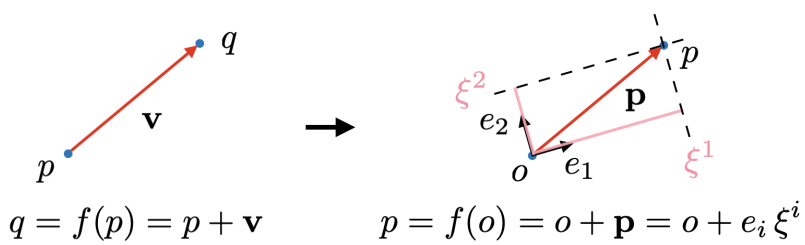

Fig. K.1: Affine spaces (A, V) and (V, V) for $A = \mathbb{R}^2 = V$.

Affine space

Given a point set A and a vector space V. For any point $a \in A$ and a vector $\mathbf{v} \in V$, we can get a new point $b \in A$ through the so-called *affine transformation*

$$\begin{aligned} f : A \times V &\longrightarrow A, \\ (a, \mathbf{v}) &\longmapsto b = f(a) = a + \mathbf{v}, \end{aligned} \tag{K.40}$$

with such a structure we call a space an *affine space* denoted by a pair (A, V). Given a point $p \in A$ we can recognized the point p as a position vector $\mathbf{p} \in V = \mathbb{R}^n$ with respect to a reference point o as an *origin* in A by using the affine transformation $p = f(o) = o + \mathbf{p}$. Let $\{e_i\}$ be a frame attached at o, the vector $\mathbf{p}$ can be written by $\mathbf{p} = e_i\, \xi^i$ with the coordinates $\{\xi^i\}$. Particularly $o = f(o) = o + \mathbf{o}$ means that the point o can be represented by a zero vector $\mathbf{o} \in V$. Therefore U is isomorphic to a vector space V, and then we obtain an affine space (V, V) such that $\mathbf{p} = \mathbf{o} + \mathbf{p}$ and $\mathbf{q} = \mathbf{p} + \mathbf{v}$ with $\mathbf{v} \in V$ corresponding to $p = o + \mathbf{p}$ and $q = p + \mathbf{v}$ in (A, V), respectively.
For $A = \mathbb{R}^2 = V$, affine spaces (A, V) and (V, V) are illustrated by Fig. K.1. With the structure of (V, V), there exist an infinitesimal displacement vector $\mathbf{v} = \mathbf{dp}$ from $\mathbf{p}$ to $\mathbf{q}$ such that $\mathbf{p}$ infinitesimally translate to $\mathbf{q} = \mathbf{p} + \mathbf{dp}$, which is shown in Fig. K.2 . Such a displacement can be described by the moving frame $\{e_i\}$ attached on p and expressed by a vector-valued (more precisely, a $\mathbb{R}^n$-valued) 1-form (a *canonical* 1-*form*) $\mathbf{dp} = \partial_\mu \otimes dx^\mu = e_i \otimes \vartheta^i := \vartheta \in A^1(E)$, which is an identity map of any vector in $\Gamma(E)$, i.e., $\mathbf{dp} \in \mathrm{End}(E)$.

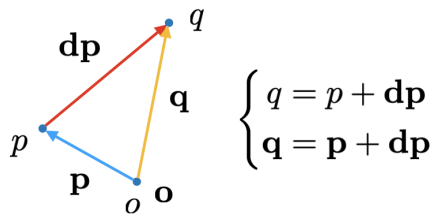

Fig. K.2: Two points p and q joining by an infinitesimal displacement vector $\mathbf{dp}$.

Local affine space. We start from a fact that the neighborhood of any point in a manifold is an local affine space. In a n-dimensional space, a neighborhood U of a point $p \in \mathcal{M}$ is regarded as an local Euclidean (resp. Minkowskian) space and it forms an affine space (U, V) with $V = T_p\mathcal{M}$ a vector space tangent to p. Let q be another point in U, which should be infinitesimally near to p. Hence, two points are joined by a displacement vector $\mathbf{dp}$ such that $q = p + \mathbf{dp}$. Because U is isomorphic to V, we can always, respectively, recognize p and q by $\mathbf{p}$ and $\mathbf{q}$ such that $\mathbf{q} = \mathbf{p} + \mathbf{dp}$ in an affine space (V, V). Every point can attach a frame. Let frames of p and q are $\{e_i\}$ and $\{e'_i\}$ respectively. Each component of two frames can be similarly identified by $e'_i = e_i + \mathbf{d}e_i$ with introducing an infinitesimal variation vector $\mathbf{d}e_i$. Both the displacement vector $\mathbf{dp}$ and the set of variation vectors $\{\mathbf{d}e_i\}$ are vectorial variations and can be expanded by the frame $\{e_i\}$ at p as

$$\boxed{\begin{cases} \mathbf{dp} = e_i \otimes \vartheta^i \overset{\text{or}}{=} e_i\, \vartheta^i \in A^1(E), & \text{(K.41a)} \\ \mathbf{d}e_i = e_j \otimes \omega^j{}_i \overset{\text{or}}{=} e_j\, \omega^j{}_i \in A^1(E). & \text{(K.41b)} \end{cases}}$$

Here the number of the introduced parameters (ϑ^i and $\omega^j{}_i$) is $n + n^2$. These two equations of (K.41) are used as a starting point to construct Cartan's structure equations below.

Cartan's first structure equation. By applying the exterior covariant differentiation on the vector-valued 1-form (K.41a), then $\mathbf{d}^2\mathbf{p}$ is

$$\begin{aligned} \mathbf{d}^2\mathbf{p} &= \mathbf{d} \circ \mathbf{dp} = \mathbf{d}\vartheta \\ &= \mathbf{d}(e_i \otimes \vartheta^i) \\ &= \nabla e_i \wedge \vartheta^i + e_i \otimes \mathrm{d}\vartheta^i \\ &= e_i \otimes (\mathrm{d}\vartheta^i + \omega^i{}_j \wedge \vartheta^j) \coloneqq e_i \otimes \Theta^i\,, \end{aligned} \tag{K.42}$$

which defines the *Cartan's first structure equation*

$$\boxed{\Theta^i = \mathrm{d}\vartheta^i + \omega^i{}_j \wedge \vartheta^j\,.}^{\dagger 24} \tag{K.43}$$

We can define $\mathcal{T} \coloneqq e_i \otimes \Theta^i \in A^2(E)$ as the torsion vector 2-form. Let γ be a *contour* around a fixed point $\mathbf{A}$, $\mathbf{p}$ and $\mathbf{p}'$ are two infinitesimal close points on γ, i.e., $\mathbf{dp} = \mathbf{p}' - \mathbf{p}$ is an infinitesimal vector along γ. We assume that $\mathbf{p}$ is the starting point and the end point is $\mathbf{q}$. As illustrated

[†24] Some people would like to define the differential operator $\mathrm{D} \sim \mathrm{d} + \omega\wedge$ on the *component* of an exterior differential form, then the torsion 2-form is written as $\Theta^i = \mathrm{D}\vartheta^i$. By Cartan's convention, it should be $\mathcal{T} = (\mathbf{dp})'$.

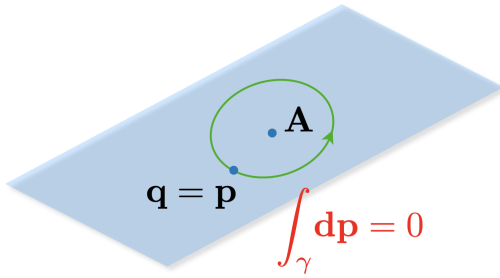

Fig. K.3: A closed contour without net displacement.

by Fig. K.3, take the sum of infinitesimal vector $\mathbf{dp}$ at every point along γ, by using the Stokes' theorem under the region D, the result would be zero:

$$0 = \mathbf{q} - \mathbf{p} = \oint_\gamma \mathbf{dp} = \int_D \mathbf{d} \circ \mathbf{dp} = \int_D e_i \otimes (\mathrm{d}\vartheta^i + \widetilde{\omega}^i{}_j \wedge \vartheta^j) = \int_D \mathcal{T}, \tag{K.44}$$

i.e., $\mathcal{T} = 0$ for a closed contour, which implies that we have chosen a particular connection $\omega^i{}_j$ such that $\mathcal{T}$ vanishes. The connection $\widetilde{\omega}^i{}_j$ is, in fact, what we called the *Riemannian connection* chosen by Riemann. This is actually the mathematical restriction on the *Rimaniann geometry*.

There are other possibilities to choose a connection such that the sum of infinitesimal vector $\int_\gamma \mathbf{dp}$ over γ is non-vanishing. If a connection $\omega^i{}_j$ is not *Riemannian* and the curve γ does not form a closed loop, it implies that a point translation is effectively associated with the *non-vanishing* torsion vector $\mathcal{T}$, which is shown in Fig. K.4. According to Cartan's first structure equation,

(a) Torsion on the surface.

(b) Torsion in the space.

Fig. K.4: Torsion is related to the translation of a point.

non-Riemannian connection leads to the non-vanishing *torsion 2-from*

$$\begin{aligned}\Theta^k &= \mathrm{d}\vartheta^k + \omega^k{}_j \wedge \vartheta^j \\ &= \Big((e_i\, e^k{}_\mu) e_j{}^\mu + \omega^k{}_{ji} \Big) \vartheta^i \wedge \vartheta^j \\ &\equiv \frac{1}{2} T^k{}_{ij}\, \vartheta^i \wedge \vartheta^j \equiv S^k{}_{ji}\, \vartheta^i \wedge \vartheta^j\,,\end{aligned} \tag{K.45}$$

where we obtain the components

$$\boxed{T^k{}_{ij} = 2\, S^k{}_{ji} = 2\, \omega^k{}_{[ji]} - c^k{}_{ij}} \tag{K.46}$$

with $c^k{}_{ij}$ the structure constants defined in (K.34).

One can define the so-called contortion 1-form

$$\boxed{K^i{}_j := K^i{}_{jk}\vartheta^k,} \quad \text{such that} \quad \boxed{T^i{}_{jk} = K^i{}_{kj} - K^i{}_{jk} = 2\,K^i{}_{[kj]},} \tag{K.47}$$

then the torsion 2-form can be written as

$$\begin{aligned}\Theta^i &= \frac{1}{2}T^i{}_{jk}\,\vartheta^j\wedge\vartheta^k = -\frac{1}{2}T^i{}_{jk}\,\vartheta^k\wedge\vartheta^j \\ &= (K^i{}_{[jk]} + K^i{}_{(jk)})\,\vartheta^k\wedge\vartheta^j = K^i{}_{jk}\,\vartheta^k\wedge\vartheta^j = \boxed{K^i{}_j\wedge\vartheta^j\,.}\end{aligned} \tag{K.48}$$

If we define another connection 1-form by

$$\varpi^i{}_j := \omega^i{}_j - K^i{}_j, \tag{K.49}$$

according to the first structure equation,

$$\begin{aligned}\mathrm{d}\vartheta^i &= -\omega^i{}_j\wedge\vartheta^j + \Theta^i \\ &= -(\varpi^i{}_j + K^i{}_j)\wedge\vartheta^j + \Theta^i \\ &= -\varpi^i{}_j\wedge\vartheta^j - \Theta^i + \Theta^i = -\varpi^i{}_j\wedge\vartheta^j\,.\end{aligned} \tag{K.50}$$

The result above shows that the 1-form $\varpi^i{}_j = \widetilde{\omega}^i{}_j$ is a *torsion-free* connection. Consequently we obtain a decomposition of a connection form

$$\boxed{\omega^i{}_j = \widetilde{\omega}^i{}_j + K^i{}_j} \tag{K.51}$$

and the corresponding components in the local coordinate system is

$$\boxed{\Gamma^\rho{}_{\nu\mu} = \left\{{\rho \atop \nu\mu}\right\} + K^\rho{}_{\nu\mu},} \tag{K.52}$$

which is the connection (**??**) with a *vanishing* nonmetricity, i.e., $Q_{\rho\nu\mu} = 0$

Cartan's second structure equation. The exterior covariant differentiation applied on the vector-valued 1-form (K.41b) leads to

$$\begin{aligned}\mathbf{d}^2 e_i &= \mathbf{d}\circ\mathbf{d}e_i \\ &= \mathbf{d}(e_j\otimes\omega^j{}_i) \\ &= \nabla e_j\wedge\omega^j{}_i + e_j\otimes\mathrm{d}\omega^j{}_i \\ &= e_j\otimes(\mathrm{d}\omega^j{}_i + \omega^j{}_k\wedge\omega^k{}_i) := e_j\otimes\Omega^j{}_i\,,\end{aligned} \tag{K.53}$$

which defines the *Cartan's second structure equation*

$$\boxed{\Omega^j{}_i = \mathrm{d}\omega^j{}_i + \omega^j{}_k\wedge\omega^k{}_i\,.}^{\dagger 25} \tag{K.54}$$

The curvature vector 2-form is defined by $\mathcal{R}_i := e_j\otimes\Omega^j{}_i \in A^2(E)$. The similar infinitesimal

[†25] Some people also define a similar differential operator $\bar{\mathrm{D}} := \mathrm{d} + \omega\wedge$ for Cartan's second structure equation analogue to $\mathrm{D} \sim \mathrm{d} + \omega\wedge$ for Cartan's first structure equation, then the curvature 2-form is written by $\Omega^j{}_i = \bar{\mathrm{D}}\omega^j{}_i$ like the expression of torsion 2-form. However, we should note that the new operator $\bar{\mathrm{D}}$ here is NOT a covariant exterior derivative because of $\bar{\mathrm{D}}\omega^j{}_i \neq \mathrm{D}\omega^j{}_i = \mathrm{d}\omega^j{}_i + \omega^j{}_k\wedge\omega^k{}_i - \omega^k{}_i\wedge\omega^j{}_k$.

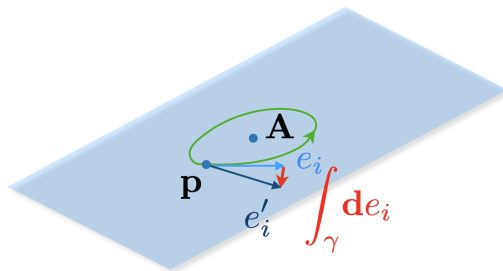

Fig. K.5: Curvature is related to the rotation of a vector.

contour integral for $\mathbf{d}e_i$ gives

$$\int_\gamma \mathbf{d}e_i = \iint_D \mathbf{d}\circ\mathbf{d}e_i = \iint_D \mathcal{R}_i\,, \tag{K.55}$$

which gives that the rotation of the basis vector is associated with the curvature. The *curvature 2-from* can be obtained by the second Cartan's structure equation

$$\begin{aligned}
\Omega^k{}_\ell &= \mathrm{d}\omega^k{}_\ell + \omega^k{}_m \wedge \omega^m{}_\ell \\
&= (\mathrm{d}\omega^k{}_{\ell m}) \wedge \vartheta^m + \omega^k{}_{\ell m} \wedge \mathrm{d}\vartheta^m + (\omega^k{}_{mi}\,\omega^m{}_{\ell j})\vartheta^i \wedge \vartheta^j \\
&= \Big(e_i\,\omega^k{}_{\ell j} + \omega^k{}_{\ell m}(e_i\,e^m{}_\mu)e_j{}^\mu + \omega^k{}_{mi}\,\omega^m{}_{\ell j}\Big)\vartheta^i \wedge \vartheta^j \\
&\equiv \frac{1}{2}\,R^k{}_{\ell ij}\,\vartheta^i \wedge \vartheta^j\,,
\end{aligned} \tag{K.56}$$

where the components are

$$\boxed{R^k{}_{\ell ij} = 2\Big(e_{[i}\,\omega^k{}_{\ell|j]} + \omega^k{}_{m[i}\,\omega^m{}_{\ell|j]}\Big) - \omega^k{}_{\ell m}\,c^m{}_{ij}\,.} \tag{K.57}$$

K.3 Bianchi Identities

Again, we directly apply the covariant exterior derivative on $\mathbf{d}^2\mathbf{p} = \mathcal{T}$ and $\mathbf{d}^2 e_i = \mathcal{R}_i$, equations are respectively read as

$$\begin{aligned}
\mathbf{d}^3\mathbf{p} = \mathbf{d}\mathcal{T} &= \mathbf{d}(e_i\,\Theta^i) = (\nabla e_i) \wedge \Theta^i + e_i(\mathrm{d}\Theta^i) \\
&= e_k\,\omega^k{}_i \wedge \Theta^i + e_i(\mathrm{d}\Theta^i) \\
&= \boxed{e_i(\omega^i{}_k \wedge \Theta^k + \mathrm{d}\Theta^i)} \\
&= e_i\big[\omega^i{}_k \wedge (\mathrm{d}\vartheta^k + \omega^k{}_j \wedge \vartheta^j) + (\mathrm{d}\omega^i{}_j \wedge \vartheta^j + (-1)^r \underbrace{\omega^i{}_j}_{r\text{-form}} \wedge \mathrm{d}\vartheta^j)\big] \\
&= e_i(\cancel{\omega^i{}_j \wedge \mathrm{d}\vartheta^j} + \omega^i{}_k \wedge \omega^k{}_j \wedge \vartheta^j + \mathrm{d}\omega^i{}_j \wedge \vartheta^j - \cancel{\omega^i{}_j \wedge \mathrm{d}\vartheta^j}) \\
&= e_i(\mathrm{d}\omega^i{}_j + \omega^i{}_k \wedge \omega^k{}_j) \wedge \vartheta^j = \boxed{e_i\,\Omega^i{}_j \wedge \vartheta^j.}
\end{aligned} \tag{K.58}$$

and

$$\mathbf{d}^3 e_i = \mathbf{d}\mathcal{R}_i = \mathbf{d}(e_j\,\Omega^j{}_i) = (\nabla e_j) \wedge \Omega^j{}_i + e_j(\mathrm{d}\Omega^j{}_i)$$

$$
\begin{aligned}
&= e_k\,\omega^k{}_j \wedge \Omega^j{}_i + e_j(\mathrm{d}\Omega^j{}_i) \\
&= \boxed{e_j(\omega^j{}_\ell \wedge \Omega^\ell{}_i + \mathrm{d}\Omega^j{}_i)} \\
&= e_j\big[\omega^j{}_\ell \wedge (\mathrm{d}\omega^\ell{}_i + \omega^\ell{}_k \wedge \omega^k{}_i) + (\mathrm{d}\omega^j{}_k \wedge \omega^k{}_i + (-1)^r \underbrace{\omega^j{}_k}_{r\text{-form}} \wedge \mathrm{d}\omega^k{}_i)\big] \\
&= e_j(\cancel{\omega^j{}_\ell \wedge \mathrm{d}\omega^\ell{}_i} + \omega^j{}_\ell \wedge \omega^\ell{}_k \wedge \omega^k{}_i + \mathrm{d}\omega^j{}_k \wedge \omega^k{}_i - \cancel{\omega^j{}_k \wedge \mathrm{d}\omega^k{}_i}) \\
&= e_j(\mathrm{d}\omega^j{}_k + \omega^j{}_\ell \wedge \omega^\ell{}_k) \wedge \omega^k{}_i = \boxed{e_j\,\Omega^j{}_k \wedge \omega^k{}_i.}
\end{aligned} \tag{K.59}
$$

Alternatively, applying the exterior derivative to the Cartan's structure equations of (K.43) and (K.54) respectively, then we have

$$
\begin{aligned}
\mathrm{d}\Theta^i &= \underbrace{\mathrm{d}^2\vartheta^i}_{0} + \mathrm{d}\omega^i{}_j \wedge \vartheta^j + (-1)^r \underbrace{\omega^i{}_j}_{r\text{-form}} \wedge \mathrm{d}\vartheta^j \\
&= (\Omega^i{}_j - \omega^i{}_k \wedge \omega^k{}_j) \wedge \vartheta^j - \omega^i{}_j \wedge (\Theta^j - \omega^j{}_k \wedge \vartheta^k) \\
&= \Omega^i{}_j \wedge \vartheta^j - \cancel{\omega^i{}_k \wedge \omega^k{}_j \wedge \vartheta^j} - \omega^i{}_j \wedge \Theta^j + \cancel{\omega^i{}_j \wedge \omega^j{}_k \wedge \vartheta^k},
\end{aligned} \tag{K.60}
$$

and

$$
\begin{aligned}
\mathrm{d}\Omega^j{}_i &= \underbrace{\mathrm{d}^2\omega^j{}_i}_{0} + \mathrm{d}\omega^j{}_k \wedge \omega^k{}_i + (-1)^r \underbrace{\omega^j{}_k}_{r\text{-form}} \wedge \mathrm{d}\omega^k{}_i \\
&= (\Omega^j{}_k - \omega^j{}_\ell \wedge \omega^\ell{}_k) \wedge \omega^k{}_i - \omega^j{}_k \wedge (\Omega^k{}_i - \omega^k{}_\ell \wedge \omega^\ell{}_i) \\
&= \Omega^j{}_k \wedge \omega^k{}_i - \cancel{\omega^j{}_\ell \wedge \omega^\ell{}_k \wedge \omega^k{}_i} - \omega^j{}_k \wedge \Omega^k{}_i + \cancel{\omega^j{}_k \wedge \omega^k{}_\ell \wedge \omega^\ell{}_i},
\end{aligned} \tag{K.61}
$$

Calculations above respectively lead to the so-called *first* and *second Bianchi identities*

$$
\boxed{\begin{cases}
\mathrm{d}\Theta^i = \Omega^i{}_j \wedge \vartheta^j - \omega^i{}_j \wedge \Theta^j \quad \Longrightarrow \quad \mathrm{d}\Theta^i + \omega^i{}_j \wedge \Theta^j = \Omega^i{}_j \wedge \vartheta^j, & \text{(K.62a)} \\
\mathrm{d}\Omega^j{}_i = \Omega^j{}_k \wedge \omega^k{}_i - \omega^j{}_k \wedge \Omega^k{}_i \quad \Longrightarrow \quad \mathrm{d}\Omega^j{}_i + \omega^j{}_k \wedge \Omega^k{}_i - \Omega^j{}_k \wedge \omega^k{}_i = 0. & \text{(K.62b)}
\end{cases}}
$$

One can define the exterior covariant derivative of the component of a vector-valued forms $A = e_i A^i$ with $A^i = A^i{}_\mu \mathrm{d}x^\mu$ denoted by D:

$$
\mathrm{D}A^i{}_\mu = \mathrm{d}A^i{}_\mu + \omega^i{}_j \wedge A^j{}_\mu \tag{K.63}
$$

Subsequently we obtain the following notation

$$
\boxed{\begin{array}{rcl}
\text{Structure equation} & & \text{Bianchi identity} \\
\Theta^i = \mathrm{D}\vartheta^i & \Longrightarrow & \mathrm{D}\Theta^i = \Omega^i{}_j \wedge \vartheta^j, \qquad \text{(K.64a)} \\
\Omega^i{}_j = \mathrm{d}\omega^i{}_j + \omega^i{}_k \wedge \omega^k{}_j & \Longrightarrow & \mathrm{D}\Omega^i{}_j = 0. \qquad \text{(K.64b)}
\end{array}}
$$

Particularly, for *Riemannian* space, we have torsion-free condition $\Theta^i = 0$, eventually the *first* Bianchi identity results in zero:

$$
\begin{aligned}
0 &= \Omega^i{}_j \wedge \vartheta^j \\
&= \frac{1}{2} R^i{}_{jk\ell}\, \vartheta^k \wedge \vartheta^\ell \wedge \vartheta^j
\end{aligned}
$$

$$= \frac{1}{2}\big(R^i{}_{jk\ell} + R^i{}_{k\ell j} + R^i{}_{\ell jk} - R^i{}_{j\ell k} - R^i{}_{kj\ell} - R^i{}_{\ell kj}\big)\vartheta^k \wedge \vartheta^\ell \wedge \vartheta^j$$
$$= 3\, R^i{}_{[jk\ell]}\, \vartheta^k \wedge \vartheta^\ell \wedge \vartheta^j, \tag{K.65}$$

i.e., in terms of the component expression, it should be reduced to the familiar form of

$$\boxed{R^i{}_{[jk\ell]} = 0.} \tag{K.66}$$

K.4 Geodesic and Autoparallel Curves

The geodesic $x^\mu_{(\mathrm{g})}(\tau)$ in U_4 is obtained by the same approach to do the variation of the arc length in Reimannian space in Sec. 7.3. It is the *extremum* of the arc length mathematically, or, physically, the *extremum* of the proper time interval:

$$\delta \int d\tau \quad \Longrightarrow \quad \frac{d^2 x^\rho_{(\mathrm{g})}}{d\tau^2} + \left\{ {\rho \atop \mu\nu} \right\} \frac{dx^\mu_{(\mathrm{g})}}{d\tau}\frac{dx^\nu_{(\mathrm{g})}}{d\tau} = 0. \tag{K.67}$$

It is apparent that the curve depends on the metric only due to the Levi–Civita connection.

Moreover, through the connection, we can define the parallel transport of any tangent vector X. If there exist a curve such that the vector $u_{(\mathrm{a})} = \frac{\mathbf{dp}_{(\mathrm{a})}}{d\tau} = e_\mu\, u^\mu_{(\mathrm{a})}$ with $u^\mu_{(\mathrm{a})} = \frac{dx^\mu_{(\mathrm{a})}}{d\tau}$ is parallel to itself

$$\frac{du^\rho_{(\mathrm{a})}}{d\tau} + \Gamma^\rho{}_{\mu\nu}\, u^\mu_{(\mathrm{a})}\, \frac{x^\nu_{(\mathrm{a})}}{d\tau} = 0 \quad \Longrightarrow \quad \frac{d^2 x^\rho_{(\mathrm{a})}}{d\tau^2} + \Gamma^\rho{}_{\mu\nu}\frac{dx^\mu_{(\mathrm{a})}}{d\tau}\frac{dx^\nu_{(\mathrm{a})}}{d\tau} = 0. \tag{K.68}$$

Here we can decompose the connection into metric and torsionic parts by $\Gamma^\rho{}_{\mu\nu} = \left\{ {\rho \atop \mu\nu} \right\} + K^\rho{}_{\mu\nu}$. Now the autoparallel curve could be written as

$$\frac{d^2 x^\rho_{(\mathrm{a})}}{d\tau^2} = -\left\{ {\rho \atop \mu\nu} \right\} \frac{dx^\mu_{(\mathrm{a})}}{d\tau}\frac{dx^\nu_{(\mathrm{a})}}{d\tau} - K^\rho{}_{\mu\nu}\frac{dx^\mu_{(\mathrm{a})}}{d\tau}\frac{dx^\nu_{(\mathrm{a})}}{d\tau}. \tag{K.69}$$

For given a local inertial coordinates $\{x^{\rho'}\}$, we would have $\left\{ {\rho' \atop \mu'\nu'} \right\}(x') = 0$. The presence of K is independent with the coordinate transformation. Then the autoparallel becomes

$$\left.\frac{d^2 x^{\rho'}_{(\mathrm{a})}}{d\tau^2}\right|_p = -K^{\rho'}{}_{\mu'\nu'}\big|_p \left.\frac{dx^{\mu'}_{(\mathrm{a})}}{d\tau}\right|_p \left.\frac{dx^{\nu'}_{(\mathrm{a})}}{d\tau}\right|_p. \tag{K.70}$$

In general, the geodesic and autoparallel curves are not the same trajectory of a particle. So we can seek the necessary condition for the equivalence between two curves. Trivially, if the connection is *torsion-free*, then we have $\Gamma^\rho{}_{\mu\nu} = \left\{ {\rho \atop \mu\nu} \right\}$ and the autoparallel is reduced to the geodesic, i.e., there is only one trajectory $x^\rho_{(\mathrm{a})} = x^\rho_{(\mathrm{g})} := x^\rho$. For non-trivial case, let $x^\rho_{(\mathrm{a})} = x^\rho_{(\mathrm{g})} := x^\rho$, by comparing geodesic and autoparallel curves, we would obtain the constrained equation

$$K^\rho{}_{\mu\nu}\frac{dx^\mu}{d\tau}\frac{dx^\nu}{d\tau} = 0. \tag{K.71}$$

By using the decomposition $K^\rho{}_{\mu\nu} = K^\rho{}_{(\mu\nu)} + K^\rho{}_{[\mu\nu]}$, (K.71) is reduced to

$$K^\rho{}_{(\mu\nu)}\frac{dx^\mu}{d\tau}\frac{dx^\nu}{d\tau} = 0 \quad \Longrightarrow \quad \boxed{K^\rho{}_{\mu\nu} = -K^\rho{}_{\nu\mu}.} \tag{K.72}$$

This condition implies the property that torsion tensor is *totally antisymmetric*:

$$-\frac{1}{2}\big(\cancel{T^{\rho}{}_{\mu\nu}} + T_{\mu}{}^{\rho}{}_{\nu} + T_{\nu}{}^{\rho}{}_{\mu}\big) = -\frac{1}{2}\big(\overbrace{-T^{\rho}{}_{\nu\mu}}^{\cancel{+T^{\rho}{}_{\mu\nu}}} - T_{\nu}{}^{\rho}{}_{\mu} - T_{\mu}{}^{\rho}{}_{\nu}\big)$$

$$\Longrightarrow \quad \boxed{T_{\mu}{}^{\rho}{}_{\nu} = -T_{\nu}{}^{\rho}{}_{\mu}} \quad \text{(antisymmetric in the “first” and “third” indices)} \tag{K.73a}$$

$$\overset{\times g_{\sigma\rho}}{\Longrightarrow} \quad T_{\mu\sigma\nu} = -T_{\nu\sigma\mu} = +T_{\nu\mu\sigma} \underset{\text{(K.73a)}}{=} -T_{\sigma\mu\nu}$$

$$\Longrightarrow \quad \boxed{T_{\mu\sigma\nu} = -T_{\sigma\mu\nu}} \quad \text{(antisymmetric in the “first” and “second” indices).} \tag{K.73b}$$